The AI Companion

A dynamic, conversational web application designed to act as an empathetic, therapeutic friend that listens to users and provides thoughtful, personalized feedback.

By: Meghana Kuruva

Under the Guidance of: Dr. Piyush Joshi

The Problem We're Solving

Mental Well-Being Crisis

- Rising levels of loneliness and mental health challenges in our digital age

Emotional Connection Gap

- Users need AI that understands human emotion, not just mechanical responses

Persistent Memory

- Building tools that remember past context and evolve with users over time

Our goal: Create a supportive companion that feels real, remembers your story, and responds with genuine empathy

Technology Architecture

Frontend Interface

- Python with Streamlit framework for clean, responsive web experience

Local Emotion Detection

- Hugging Face Transformers for instant emotion classification

Memory Database

- SQLite for storing context, preferences, and conversation histor

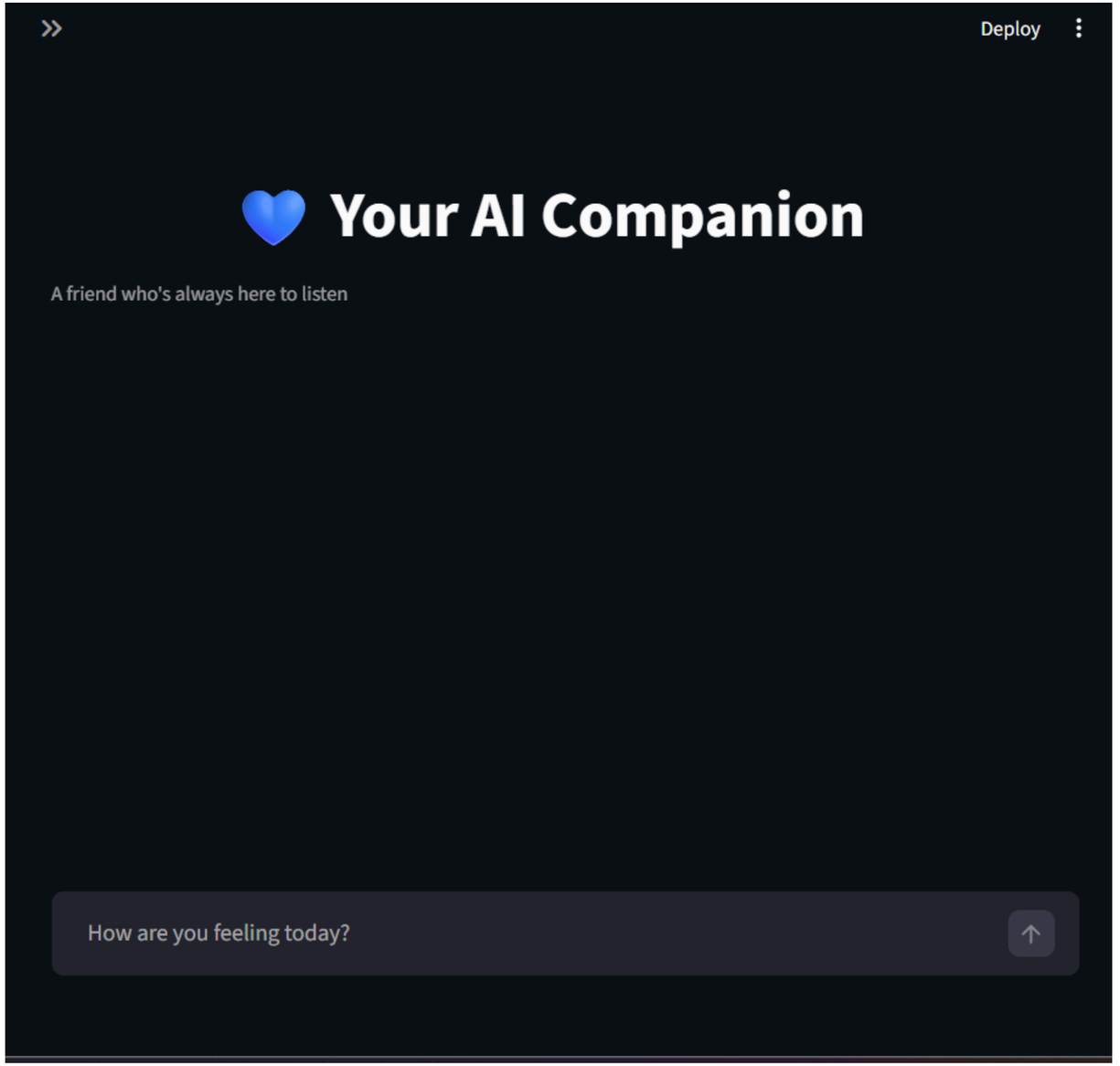
Generative AI Brain

- Google Gemini 2.5-flash for human-like conversational responses

Processing Pipeline

Step 1: Emotion Detection

When a user types a message, it's instantly sent to our local Emotion Classification model using the robertabase-go_emotions pre-trained model. The AI tags the text with primary emotions4joy, sadness, anger, anxiety4 and assigns confidence scores. This processing happens locally and securely



Step 2: Memory Extraction

Text Analysis

- Regex and keyword tracking extract specific feelings and meaningful details from user messages

Memory Storage

- Key information like "I enjoy basketball" gets saved as 'liked' memories into the SQLite database

Context Building

- Database stores recent conversation history, creating persistence across sessions

Step 3: Chain of Thought

We combine all extracted information into a rich contextual prompt that gives Gemini deep psychological insight into the user's state:

Deep Emotion -Emotion tags from local AI analysis

Explicit Feelings-Directly mentioned emotions and states

Extracted Memories-Stored preferences and personal details

Conversation History-Recent dialogue context and flow

Confidence Scores & Reinforcement Learning Strategy

- Confidence Scores When the roberta-basego_emotions model analyzes user text, it doesn't just output one emotion. Instead, it generates a Probability Distribution across all 28 emotion categories. The Confidence Score is a percentage (0.95 = 95%) representing mathematical certainty. For example, if confidence is 0.95 for "Anger", the neural network is 95% certain that anger is the dominant emotion. This score is critical because it tells Gemini exactly how intense the user's emotion is, enabling more nuanced and appropriate responses.

- Why Reinforcement Learning While the emotion detection and memory extraction are excellent, the generative responses from Gemini can still be improved. RLHF allows the system to learn from actual user feedback. Users rate responses as "Helpful" or "Not Helpful", creating a feedback loop that teaches the AI what kinds of responses are truly empathetic and effective for different emotional states.

- Where RLHF is Applied RLHF is applied specifically to finetune the Gemini model's response generation. By collecting thousands of user feedback ratings alongside the exact emotional context (emotion tags, confidence scores, memories, conversation history), developers can train reward models that learn to predict which responses users find most helpful. These reward models then guide the fine-tuning of Gemini to generate increasingly empathetic and contextually appropriate responses.

Research Foundation & Methodologies

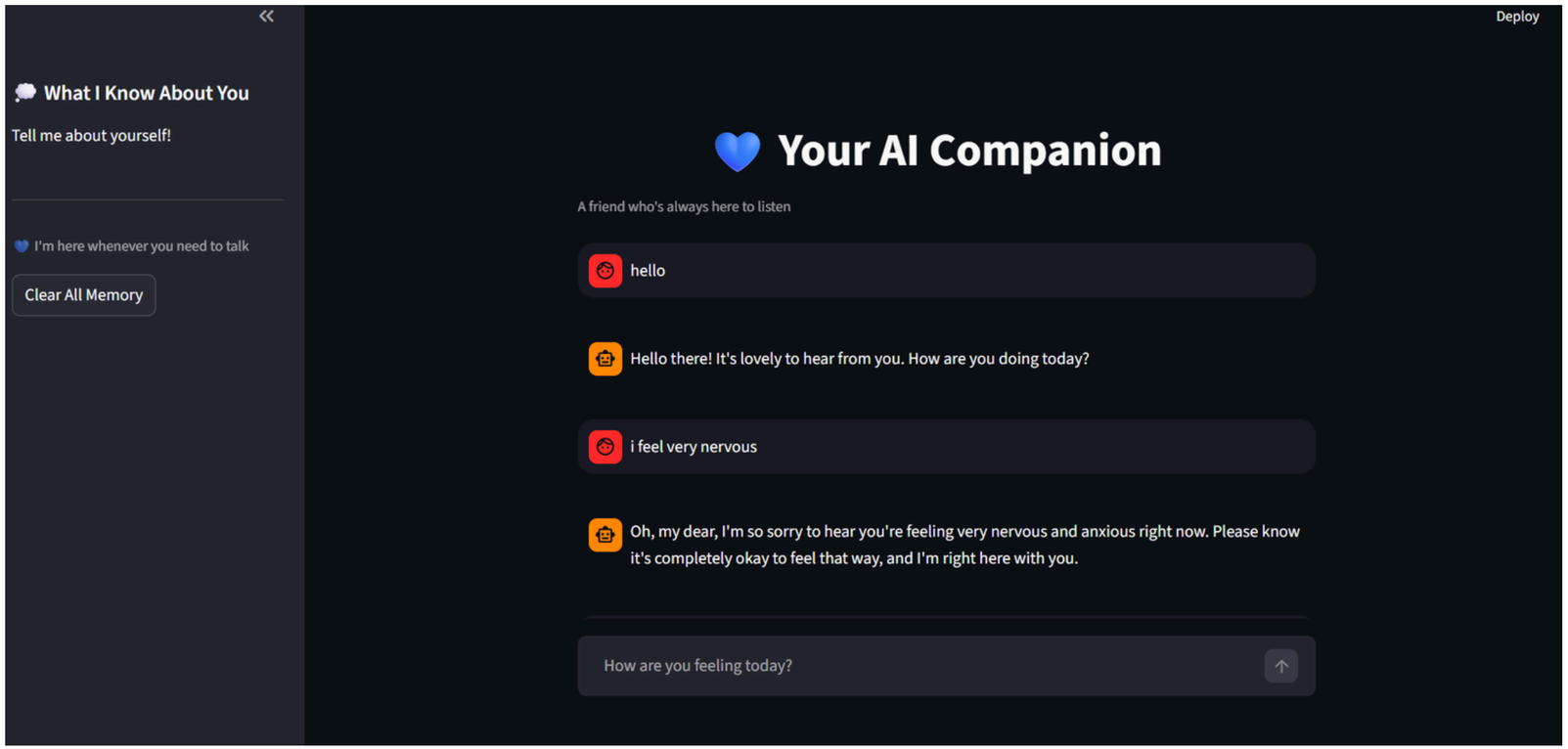
Chain-of-Thought Distillation (Interspeech 2025)
Method: Uses AECoTD (Acoustic Emotional Chain-of-Thought Distillation) with DeepSeek-R1 for reasoning and Qwen-2.5-7B for distillation.
Approach: Performs acoustic analysis (pitch, intensity) before inferring emotion, building reasoning paths that are distilled into smaller models.

OmniVox: Zero-Shot Emotion Recognition (arXiv 2025)
Method: Compares "Minimal" vs. "CoT" prompting strategies with Gemini and GPT-4o. **Approach**: Explicitly instructs models to reason through context and acoustic features, significantly boosting zero-shot performance in Emotion Recognition in Conversation (ERC).

CoE: Clue of Emotion Framework (ACL 2025)
Method: Multi-stage learning strategy using auxiliary tasks.
Approach: Uses CoT-style tasks like role-playing, speaker identification, and rational reasoning to help LLMs understand the "why" behind emotions, identifying non-obvious secondary emotions.

Step 4: Response Generation

The enriched Chain of Thought prompt is securely sent through the Google Gemini API. By providing emotion tags, context, and memories alongside the prompt, we guarantee responses that are highly empathetic, humanlike, and directly contextual to the user's immediate mood state



User Interface & Dashboard

- Clean Chat Interface Modern, text-focused design resembling contemporary messaging applications
- Persistent Sidebar Displays remembered information about the user, reinforcing personal connection
- Feedback System Like/dislike buttons save user preferences for future finetuning

Key Benefits

- Genuine Empathy Understands emotional context and responds with appropriate sensitivity
- Personalized Experience Remembers likes, dislikes, and conversation history across sessions
- Local Processing Emotion detection and memory storage happen securely on your device

Future Roadmap

- **Voice-to-Text Inputs Expand accessibility with natural voice conversations**
- **Emotional Check-In Graphs Visual tracking of mood patterns over time**
- **Enhanced Personalization Deeper learning from user feedback and interactions**

By combining lightweight local AI with advanced generative models, we've created a deeply personal therapeutic companion that evolves into a true friend.

Thankyou